

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

Robotics and Computer Vision Engineer with 3+ years' experience developing real-time perception and SLAM systems for safety-critical industrial environments. Skilled in designing and deploying computer vision and ROS2-based localisation and mapping pipelines using RTAB-Map and slam_toolbox. Experienced in optimising deep learning and SLAM performance on edge hardware with OpenVINO, TensorRT across Intel VPUs and Nvidia Jetson AGX Xavier. Adept at full-lifecycle delivery from data processing and model optimisation to on-site deployment and validation in harsh, dynamic conditions.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Led strategic MLOps and optimisation initiatives using Bayesian hyperparameter tuning to enhance CNN and time-series model performance across deployments.
- Integrated AI modules with production systems (Docker, FastAPI, Kafka, Git CI/CD) in collaboration with software/hardware/QA teams to deliver fault-tolerant edge deployments.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Oversaw on-site deployment of the system at a British Petroleum oil rig in the Omani desert.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing throughput by 233% and enabling real-time detection of safety-critical events.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50%.
- Mentored engineers on optimisation and deployment practices, raising code quality and delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for teleoperation of an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimization.
- Tuned SLAM parameters to improve localisation accuracy across varied terrains, achieving stable performance in both feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.
- Implemented waypoint navigation in simulation using ROS2 Nav2 (standard planners/controllers) within Gazebo, validating end-to-end navigation workflows.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Perception & SLAM:** Vision Transformer (ViT), OpenCV, LiDAR SLAM, Open3D, Point Cloud Library, 3D Feature Extraction, Scan Matching, Denoising, NumPy, Scikit-Image, Data Augmentation
- **Deep Learning and Optimisation:** OpenVINO, TensorRT, ONNX, Convolutional Neural Networks, Optuna, MultiRocket, DeepSORT, Bayesian Optimisation, PyTorch, TensorFlow, Quantisation-Aware Training
- **Programming & Development:** Python, C++, Bash, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), Pandas, PostgreSQL, Gazebo (Harmonic), Nav2
- **Cloud, Edge & MLOps:** AWS, Microsoft Azure, Intel VPUs, NVIDIA Jetson, FastAPI, Apache Kafka, MLFlow, DVC (Data Version Control), Docker, Linux, Git